Diyuan Deng

B.Sc. candidate, Data Science & Big Data Technology | LLM systems, embodied AI, applied ML

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GitHub

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) *Shenzhen, China*

B.Sc., Data Science and Big Data Technology *Sept. 2023 – Jun. 2027 (expected)*

- GPA: **3.43** / 4.0.
- Coursework: Database Systems, Algorithm Design and Analysis, Stochastic Processes, Optimization, Machine Learning, Statistical Computing.

Experience

Data Engineering Intern Shanghai Wuzhi Evolution Technology Co., Ltd. *Shanghai | Jun. 2025 – Aug. 2025*

Embodied Intelligence Data Research Dept. Collaborated with Prof. Bo Zhao's group (Shanghai Jiao Tong University) on embodied AI data and VLA models.

- Built an **xArm** robotic-arm **data acquisition pipeline** and contributed to **VLA** model optimization; led collection of **10,000+** high-quality manipulation trajectories.
- Adapted the VLA stack to the **LeRobot** framework for end-to-end general task execution; integrated multimodal LLMs and a **natural-language command interface** to lower the human–robot interaction bar.
- On hard composite tasks (e.g. autonomous grasping and multi-layer stacking), raised **task success rate above 90%**.

Selected Projects

Core Developer — Enterprise LLM Inference (SFGlang) SF Technology Co., Ltd. *Jan. 2026 – Present*

- Addressed throughput and communication bottlenecks when serving **Qwen3**-class LLMs and multimodal models (**4B / 30B**) under high enterprise concurrency.
- Stood up an automated benchmarking stack on **SGLang** with a **dual-container isolation** design on an **RTX 4090** cluster; tuned **PD/EPD disaggregation**, **TP/DP/EP** parallelism, and **FP8/INT8** quantization; shipped a one-stop **load-testing web platform**. Co-authored **two patent applications** from the core optimization work.
- Codified a deployment rule of thumb: *quantize to slim weights first, minimize TP, maximize DP*. For **Qwen-30B**, a **TP4 + DP2** schedule cut **time-to-first-token (TTFT) by 48.9%** and lifted output throughput by **664%**; for **Qwen-4B**, **TP1 + DP2** improved throughput by **750%**.

NSC-Advisor — Ischemic Stroke Prognosis (Binary: favorable vs. unfavorable) Research Assistant *Sept. 2025 – Dec. 2025*

Related work submitted to **ISMRM** (under review).

- Built a high-accuracy, interpretable prognosis system from admission-time clinical data using **few-shot learning**, prompt engineering, and a **RAG** layer over external medical evidence to resolve conflicting records.
- Helped deliver a dual-input frontend prototype with visualized outputs. Final model reached **93.75%** accuracy on **16** held-out real patient cases with improved stability and decision reliability.

Library Management System *Sept. 2024 – Nov. 2024*

- Full-stack **Python (tkinter) + SQLite**: four normalized tables, composite indexes for hot queries, **15+** features (lending/reservations, **SHA-256** authentication), admin dashboard; tests covered **100%** of specified requirements.

2D RPG Combat Prototype (UE5 + C++) *Sept. 2024 – Dec. 2024*

- Implemented combat in **C++** (three enemy AI archetypes, dynamic levels); **Qt** launcher with **SHA-256**-secured messaging; **PaperZD** animation state machines with hit detection and cooldowns. Team contributed **1,000+** lines of core code; project named **best course case**.

Leadership, Teaching & Campus Roles

Undergraduate Student Teaching Fellow (USTF) School of Data Science, CUHK-Shenzhen *Sept. 2024 – Present*

- Three consecutive semesters teaching **Data Structures**, **Databases**, and one additional core course; delivered **10+** tutorials to **800+** students; co-designed problems and grading. **SDS Excellent USTF Award (2024–2025)**.

Founder & Vice President 2Tired Cycling Club, CUHK-Shenzhen *Jul. 2024 – Present*

- Built the club from zero: **20**-person core team across five functions; organized large rides (e.g. Songshan Lake) and a **Tour de France** athlete meet-up; first-year membership **80+**, event satisfaction **95%**.

Student Assistant University Library, CUHK-Shenzhen *Nov. 2023 – Present*

- Ran **Bambu X1E**, **Ultimaker S5**, and **Form 2** industrial printers end-to-end (slicing, supports, finishing); **30+** jobs. Delivered **10+** edits (**Final Cut Pro**, **Photoshop**); led 3D-printing workshops on slicing.

Honors & Awards

- SDS Excellent USTF Award, School of Data Science (2024–2025).
- Third Prize, Guangdong Division, China Undergraduate Mathematical Contest in Modeling (2025).
- Bronze, Men's 1500 m — Inaugural CUHK-Shenzhen University Games.
- Bronze, Men's 100 m breaststroke — 7th CUHK-Shenzhen Sports Festival.

Skills & Interests

Languages	Chinese (native); English (professional proficiency for coursework and work).
Technical	**Python** (advanced), **SQL**; C++, R; **Unreal Engine 5**; ML/LLM serving (SGLang, quantization, parallelism); data engineering; GUI (tkinter/Qt).
Interests	Cycling, photography, hiking, amateur radio, Model United Nations.